AQA 3.1.8 Thermodynamics: lattice enthalpies

AQA A-level Chemistry · Revision questions · Sheet 2 of 3

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 2 of 3

22 marks · about 25 minutes

Key information

- Born–Haber value bigger than perfect ionic value → covalent character (small, highly charged cation polarises the anion)
- Second electron affinity of oxygen is endothermic: O^- repels the incoming electron
- Lattice formation: $M^+(g) + X^-(g) \rightarrow MX(s)$, exothermic; lattice dissociation is the reverse and endothermic
- Larger lattice enthalpy: higher ionic charges and smaller ions give stronger attraction
- Perfect ionic model: ions are perfect spheres / point charges with no covalent character

Section A: Short-answer questions

01 State the meaning of the term *enthalpy change*.

[1 mark]

02 Define the term *enthalpy of lattice dissociation*.

[2 marks]

03 Lattice enthalpy values can be obtained from Born–Haber cycles and by calculations based on a perfect ionic model. Which compound shows the greatest percentage difference between these two values?

[1 mark]

A CsF
B CsI
C LiF
D LiI

04 Define the term *enthalpy of atomisation* of an element.

[2 marks]

05 Give an equation, including state symbols, for the process that represents the enthalpy of lattice formation of magnesium chloride.

[1 mark]

06 Give an equation, including state symbols, for the process that represents the first electron affinity of chlorine.

[1 mark]

07 Give an equation, including state symbols, for the process that represents the enthalpy of atomisation of chlorine.

[1 mark]

08 Explain why the enthalpy of lattice formation of sodium fluoride is more exothermic than that of sodium chloride.

[2 marks]

Section B: Longer questions

09 The table shows enthalpies of lattice dissociation from Born–Haber cycles and from a perfect ionic model for three compounds.

Enthalpy of lattice dissociation / kJ mol^{-1}	Born–Haber cycle	Perfect ionic model
Sodium bromide	+747	+735
Silver bromide	+903	+759
Magnesium oxide	+3791	+3795

09.1 State **two** assumptions made in the perfect ionic model.

[2 marks]

09.2 Identify the compound with the most covalent character. Explain your answer.

[3 marks]

09.3 Explain why the values for magnesium oxide are much larger than those for sodium bromide.

[2 marks]

10 This question is about enthalpies of lattice dissociation.

10.1 Theoretical values for enthalpies of lattice dissociation can be calculated using a perfect ionic model. State the meaning of the term *perfect ionic model*.

[1 mark]

10.2 Explain why the second electron affinity of oxygen has a positive value.

[1 mark]

10.3 Explain why the enthalpy of lattice dissociation for sodium oxide is greater than the enthalpy of lattice dissociation for sodium chloride.

[2 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Sheet 2 of 3

Question	Marking guidance	Comments	Mark
01	Heat (energy) change at constant pressure	*Source: 7405/1 Nov 2021 Q01.1* Ignore energy change	1
02	M1 the enthalpy change when one mole of a (solid) ionic compound M2 dissociates (fully) into gaseous ions	*Source: 7405/3 Jun 2022 Q01.1* Ignore standard states / conditions M2 allow suitable equation with state symbols; not one mole of gaseous ions	2
03	**D**	*Source: 7405/3 Jun 2018 Q33*	1
04	M1 enthalpy change when one mole of gaseous atoms is formed M2 from the element in its standard state	*Source: Practice question (written for this set)*	2
05	$Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s)$	*Source: Practice question (written for this set)*	1
06	$Cl(g) + e^- \rightarrow Cl^-(g)$	*Source: Practice question (written for this set)*	1
07	$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	*Source: Practice question (written for this set)* Not $Cl_2 \rightarrow 2Cl$	1
08	M1 F^- is smaller than Cl^- / has a higher charge density M2 stronger attraction between the oppositely charged ions	*Source: Practice question (written for this set)* Not atoms; ignore electronegativity	2
09.1	ions are perfect spheres / point charges no covalent character / no polarisation of ions	*Source: Practice question (written for this set)*	2
09.2	M1 silver bromide M2 largest difference between the Born–Haber and perfect ionic model values M3 Ag^+ polarises / distorts the Br^- electron cloud		3
09.3	M1 Mg^{2+} and O^{2-} have higher charges / higher charge densities M2 stronger attraction between the ions		2
10.1	(Ions are) point charges / perfect spheres / no covalent character	*Source: 7405/1 Jun 2024 Q08.1, 08.4, 08.5* Do not accept atoms or molecules; allow no polarisation of ions	1
10.2	O^- repels the electron (being added)	Allow negative ion repels electron	1
10.3	M1 oxide ions have higher charge / smaller size / higher charge density (than chloride ions) M2 stronger attraction between (O^{2-} and Na^+) oppositely charged ions	Ignore electronegativity	2